Autonomy of Unmanned Aerial Vehicles

Introduction to Inverted Pendulum
Part B. Regulator Control Design

by

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Tecnológico de Monterrey

Oct 3rd, 2024

Context

It is an advanced level course in control engineering, in which the student implements their own control algorithms, applying the competencies developed in the focus stage of their study plan.

- It requires prior knowledge of modeling and automation and control system design.
- The student supports his proposal by the analysis of the performance of the control system.

QDrone 2 | Qube-Servo 3 | Aero 2

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Configuration 1

Qube-Servo 3

Configuration 2

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DC Motor

It is supplied with two standard items:

- Inverted pendulum

  The inverted pendulum is mounted to the equipment with magnets. Here, the encoder provide an additional sensor input to the system [6].

- Inertia disk

  A small aluminum part is mounted to the equipment using magnets.

Qube-Servo 3

Fig. 1 Inverted pendulum.

Fig. 2 Inertia disk.

Objective: To design a regulator control to preserve the vertical position of the rotatory inverted pendulum using Real-Time Hardware-in-the-Loop Implementation by Quanser Qube Servo 3.

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Abstract

One of the major concerns in control engineering is related to stability. If the system is unstable, we need a state feedback controller to stabilize it. Even if the system is stable, we may still regulate it according some design specifications. It is possible if the it is controllable, meaning that for any initial state vector x_a and any desired final state x_b, there is a control input $u(t)$ that can steer the state of the system from x_a to x_b in finite time. In this lab, you will design a regulator control to preserve the vertical position of the rotatory inverted pendulum.

Topics Covered

- Controllability
- State feedback control
- Companion matrices
- Pole placement control design
- Control design for higher order systems

Prerequisites

- Hardware Interfacing laboratory experiment
- Rotary Pendulum Modeling laboratory experiment

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The rotary pendulum is shown in Fig. 1.1. The model was presented in Fig. 1.2. Now, the new model is shown in Fig. 1.3:

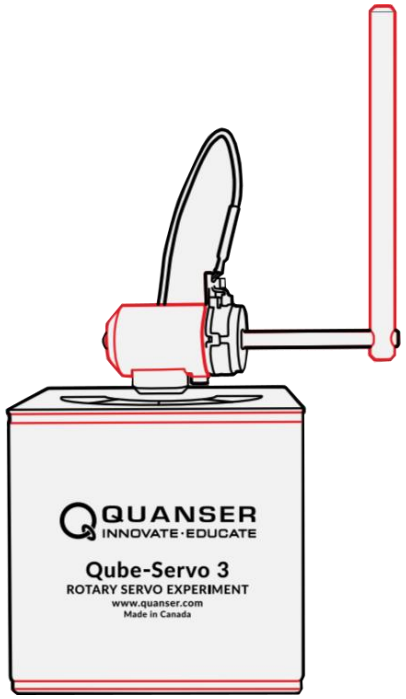

Fig. 1.1 Qube-Servo 3 Inverted pendulum.

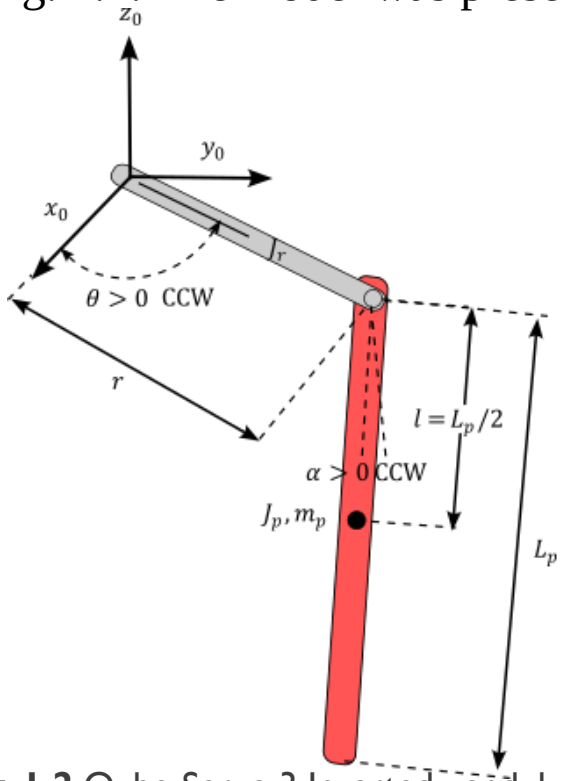

Fig. 1.2 Qube-Servo 3 Inverted pendulum DOWN / Crane Configuration.

Considering:

- The rotary arm pivot is attached to the Qube Servo 3. The arm has:
  - a length of r,
  - a moment of inertia of J_r,
- Its angle $\theta(t)$ increases positively when it rotates counterclockwise (CCW).
- The servo (and thus the arm) should turn in the CCW direction when the control voltage is positive, $v_m(t) > 0$.
- The pendulum link is connected to the end of the rotary arm. It has:
  - a total length of L_p ,
  - the center of mass is at $l = L_p/2$,
  - the moment of inertia about its center of mass is J_p,
- The rotary pendulum angle $\alpha(t)$ is zero when it is hanging downward and increases positively when rotated CCW.

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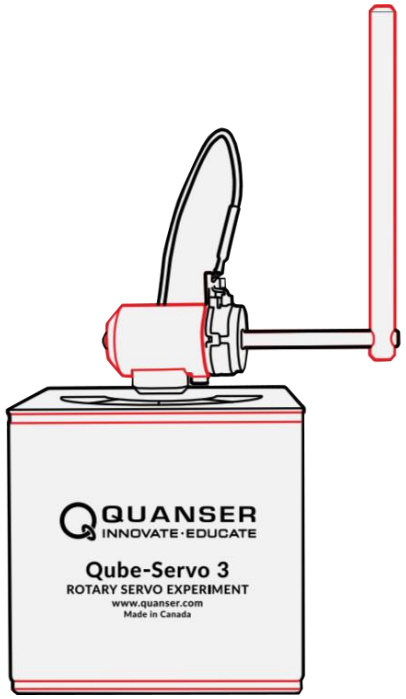

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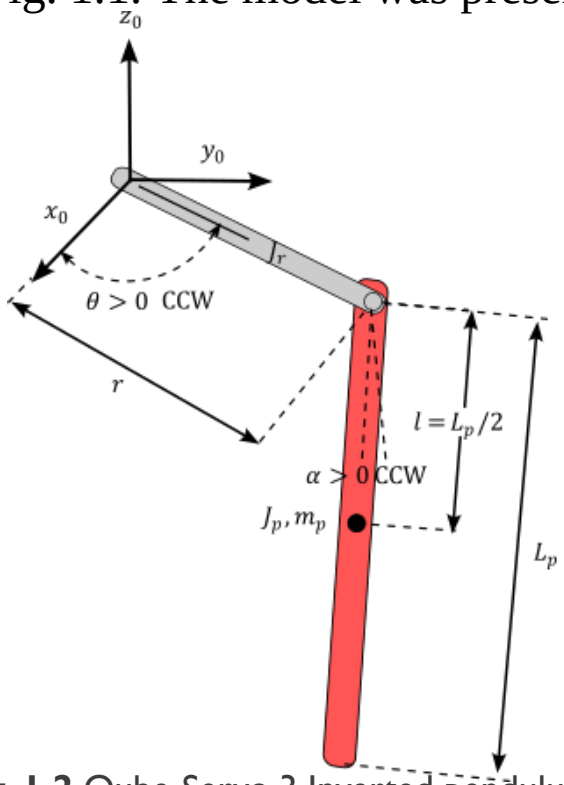

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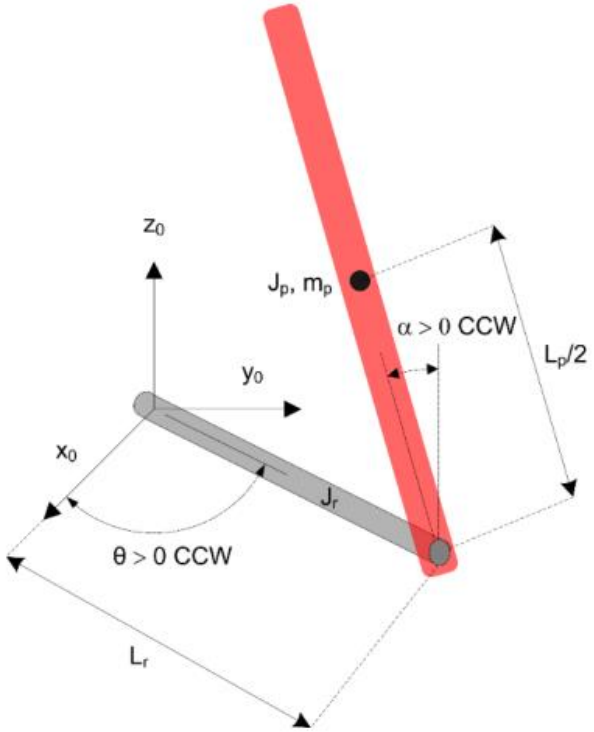

Fig. 1.3 Qube-Servo 3 Rotatory Inverted pendulum Configuration.

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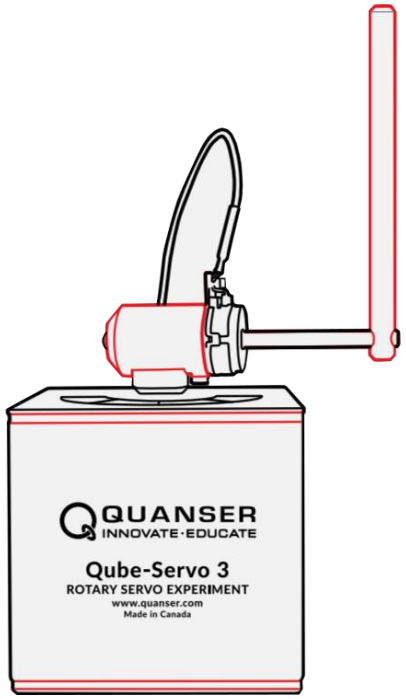
Example: Inverted Pendulum



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1. Analyze the differential equation that describes the process behavior of the system.
2. Obtain the linear time invariant state space representation.
3. Design a control regulator to achieve the following time domain specifications when the pendulum is balanced and tracking a rotary arm setpoint.

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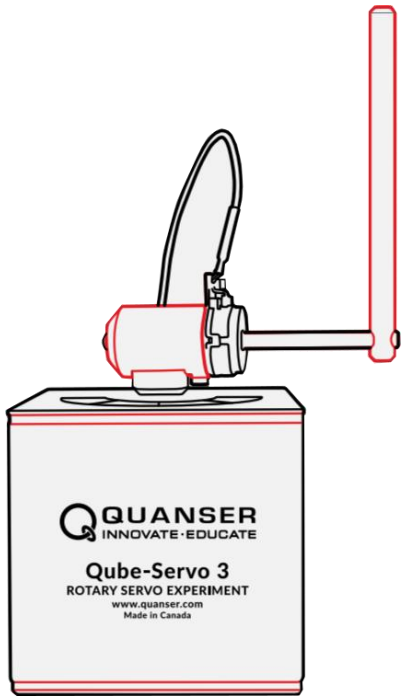

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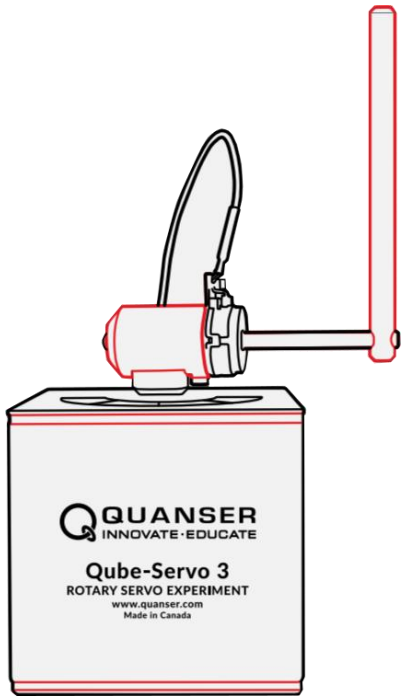

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5. Verify if the system is controllable using the controllability matrix.
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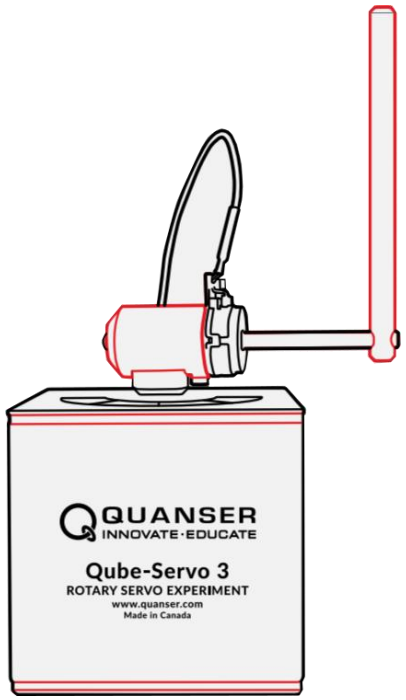

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1. Analyze the differential equation that describes the process behavior of the system.

The resultant linear equations of motion (EOM) in DOWN / Crane Configuration were developed using the Euler Lagrange method. For Rotatory Inverted pendulum Configuration is:

$$J_r\ddot{\theta}(t) + m_p l r \ddot{\alpha}(t) = \tau - b_r\dot{\theta}(t) \qquad \tau = \frac{k_m}{R_m}\left[v_m(t) - k_m\dot{\theta}(t)\right]$$

$$J_p\ddot{\alpha}(t) + m_p l r \ddot{\theta}(t) + m_p g l \alpha(t) = -b_p\dot{\alpha}(t)$$

where:

- $J_r = m_r r^2/3$ is the moment of inertia of the rotary arm with respect to the pivot (i.e. rotary arm axis of rotation)
- $J_p = m_p {L_p}^2/3$ corresponds to the moment of inertia of the pendulum link relative to the pendulum pivot (i.e. axis of rotation of pendulum)
- b_r represents the viscous damping acting on the rotary arm
- b_p stands for the viscous damping acting on the pendulum link
- τ is the applied torque at the base of the rotary arm

2. Obtain the linear time invariant state space representation.

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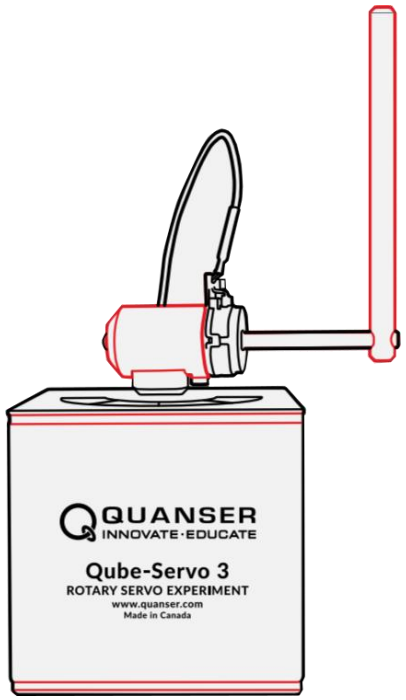

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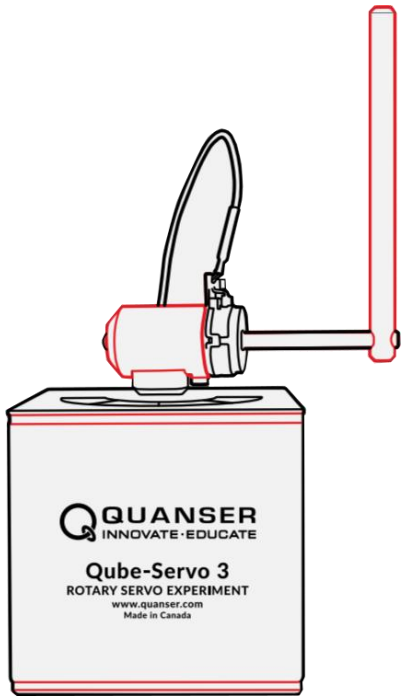

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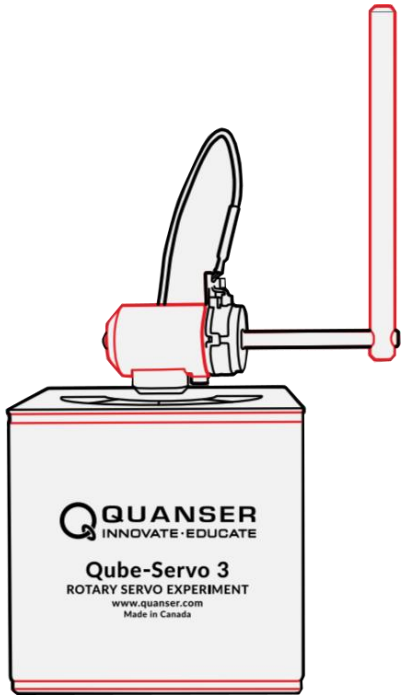

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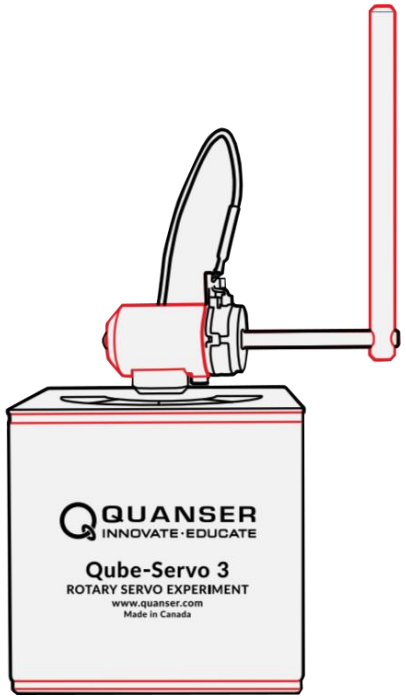

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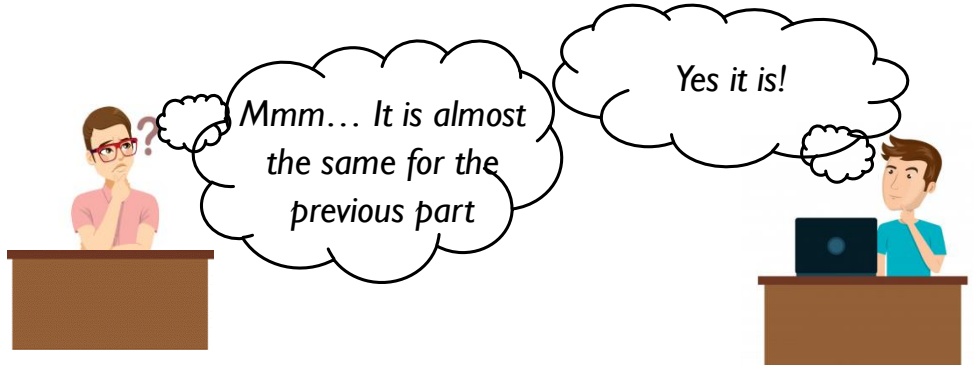

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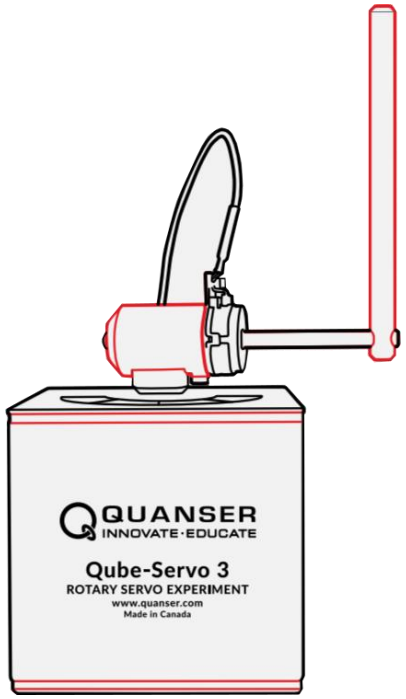

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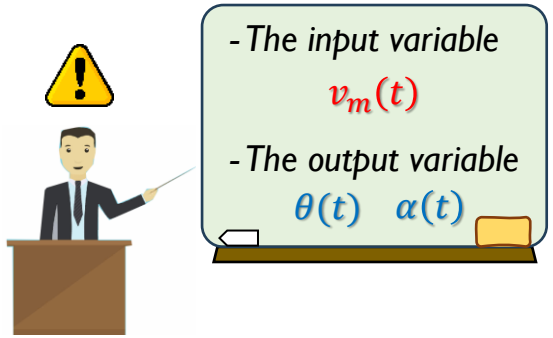

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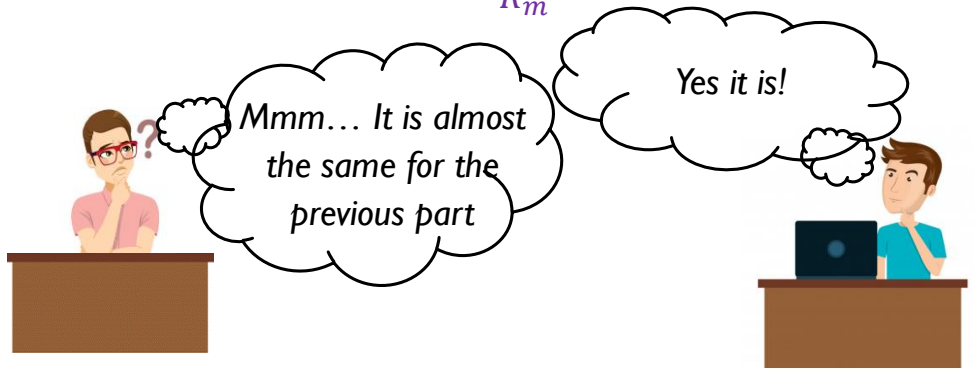

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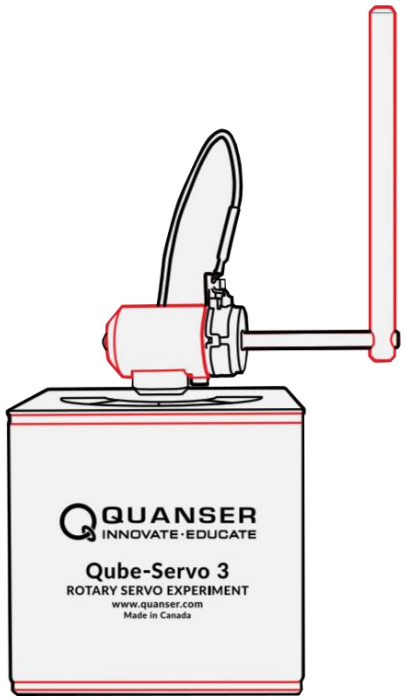

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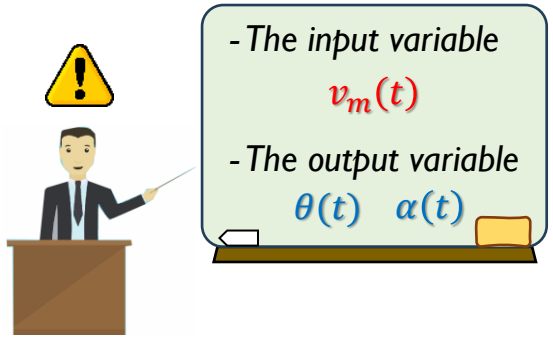

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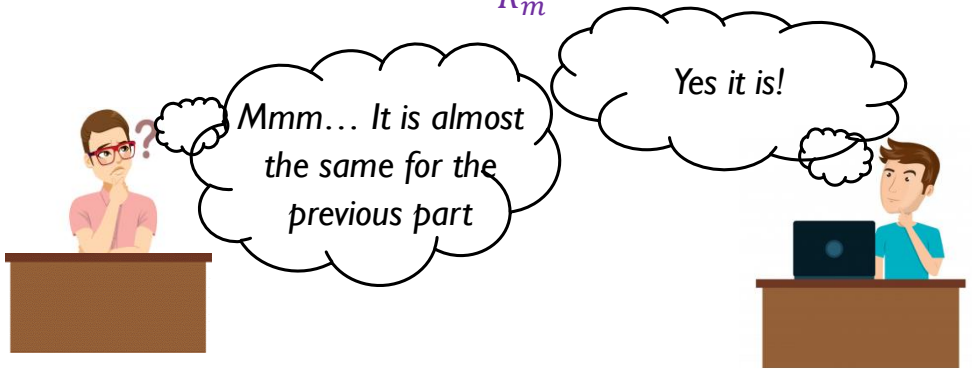

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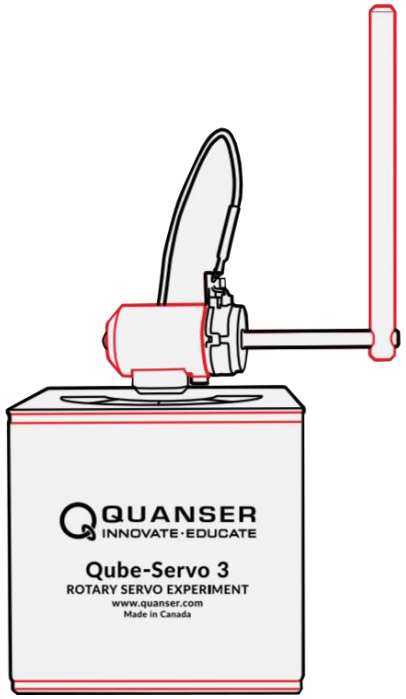

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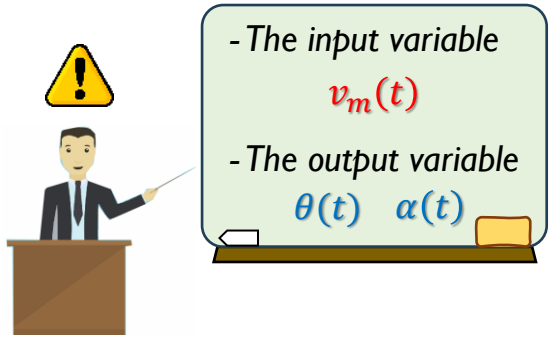

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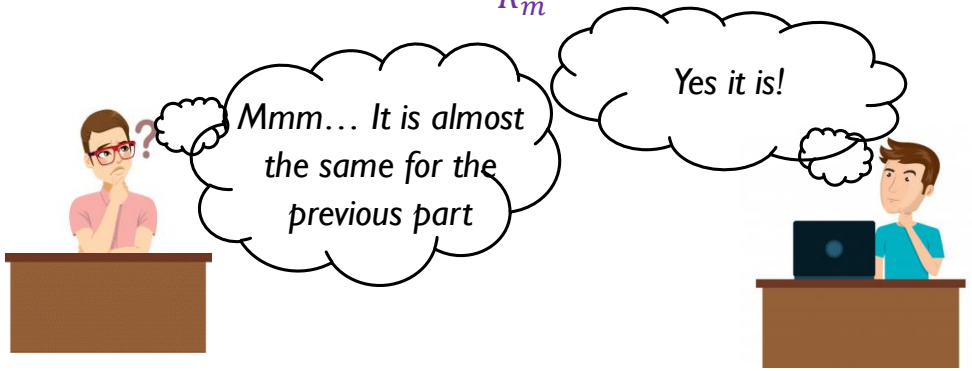

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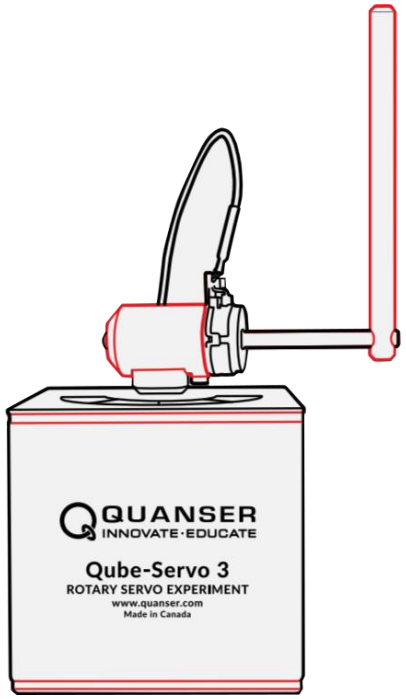

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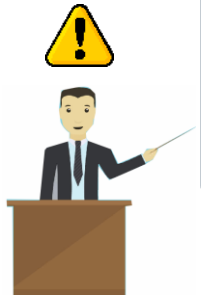

- *The input variable*

$v_m(t)$

- *The output variable*

$\theta(t) \quad \alpha(t)$

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$$\dot{\boldsymbol{x}}(t) = \mathbf{A}\boldsymbol{x}(t) + \mathbf{B}\boldsymbol{u}(t)$$

$$\boldsymbol{y}(t) = \mathbf{C}\boldsymbol{x}(t) + \mathbf{D}\boldsymbol{u}(t)$$

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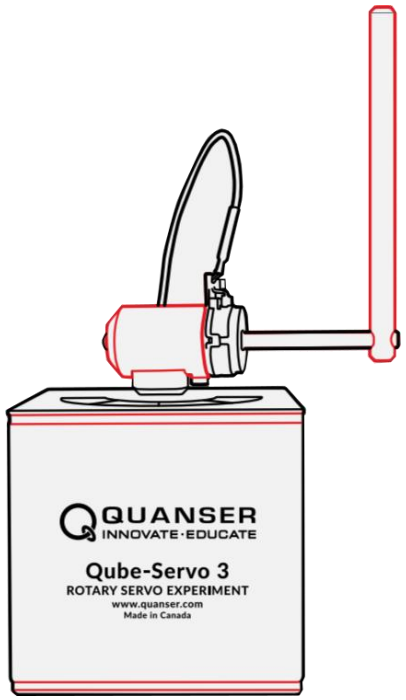

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$$\begin{bmatrix}\dot{x}_1(t)\\ \dot{x}_2(t)\\ \dot{x}_3(t)\\ \dot{x}_4(t)\end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \dfrac{m_p^2 l^2 r g}{J_t} & -\dfrac{J_p b_r}{J_t} - \dfrac{J_p \frac{{k_m}^2}{R_m}}{J_t} & +\dfrac{m_p l r b_p}{J_t} \\ 0 & -\dfrac{m_p g l J_r}{J_t} & +\dfrac{m_p l r b_r}{J_t} + \dfrac{m_p r l \frac{{k_m}^2}{R_m}}{J_t} & -\dfrac{J_p b_p}{J_t} \end{bmatrix} \begin{bmatrix}x_1(t)\\ x_2(t)\\ x_3(t)\\ x_4(t)\end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dfrac{J_p k_m}{J_t R_m} \\ -\dfrac{m_p r l k_m}{J_t R_m} \end{bmatrix} [v_m(t)]$$

$$\begin{bmatrix}y_1(t)\\ y_2(t)\end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix}x_1(t)\\ x_2(t)\\ x_3(t)\\ x_4(t)\end{bmatrix} + \begin{bmatrix}0\\ 0\end{bmatrix} [v_m(t)]$$

Practice 09

Part B. Regulator Control Design

Example: Inverted Pendulum

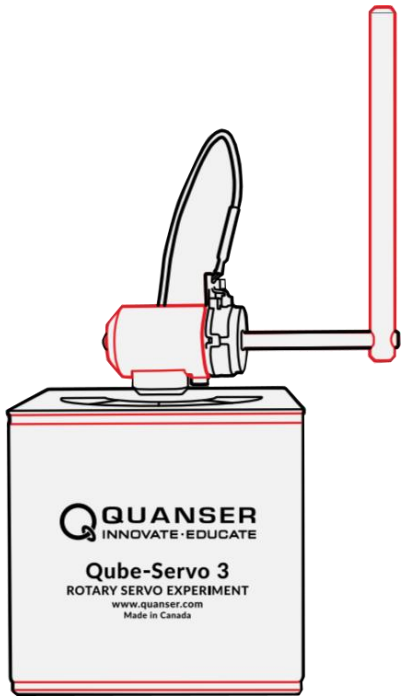

Fig. 1.1 Qube-Servo 3 Inverted pendulum.

1. Analyze the differential equation that describes the process behavior of the system.
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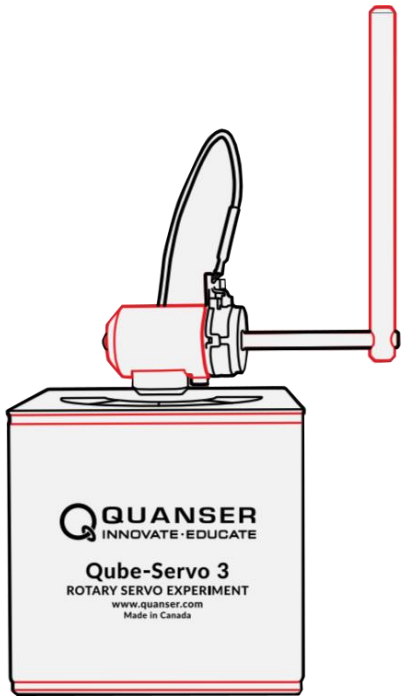
Part B. Regulator Control Design

Example: Inverted Pendulum



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Practice 09

Part B. Regulator Control Design

Example: Inverted Pendulum

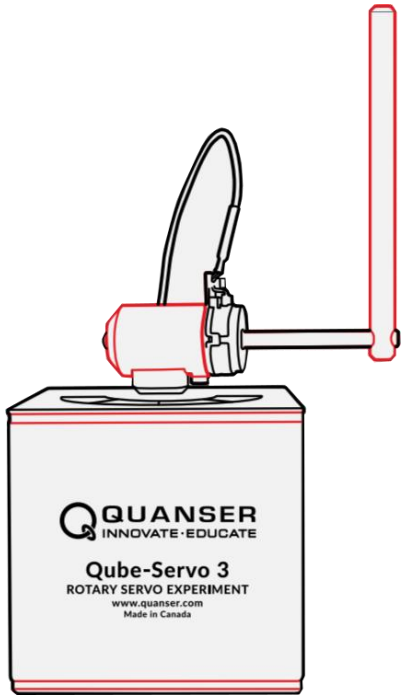

Fig. 1.1 Qube-Servo 3 Inverted pendulum.

1. Analyze the differential equation that describes the process behavior of the system.
2. Obtain the linear time invariant state space representation.
3. Design a control regulator to achieve the following time domain specifications when the pendulum is balanced and tracking a rotary arm setpoint. $\%MP = 6.81\%$ $t_{ss} = 1.54$ [sec]
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5. Verify if the system is controllable using the controllability matrix.
6. Obtain the state feedback gain matrix **K**.

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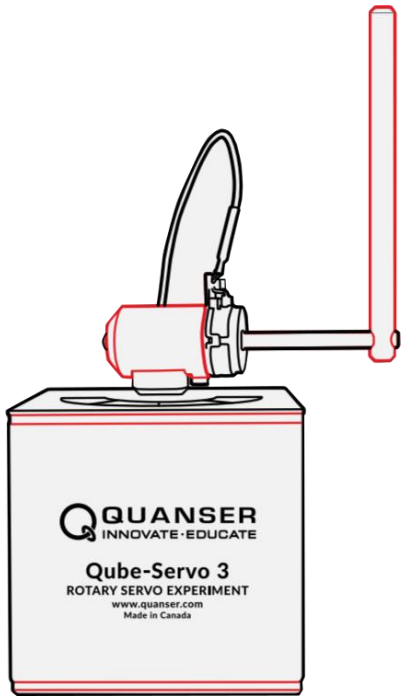
Practice 09

Part B. Regulator Control Design

Example: Inverted Pendulum



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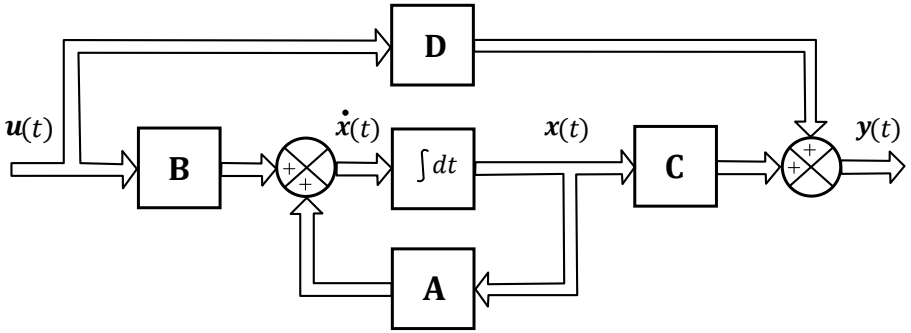

Fig. 1.4 Block Diagram.

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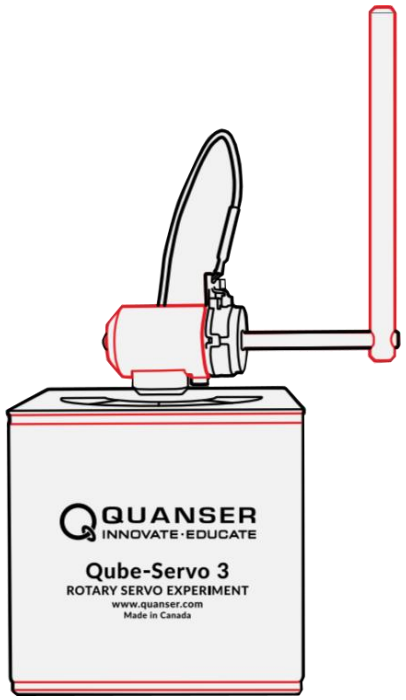
Part B. Regulator Control Design

Example: Inverted Pendulum



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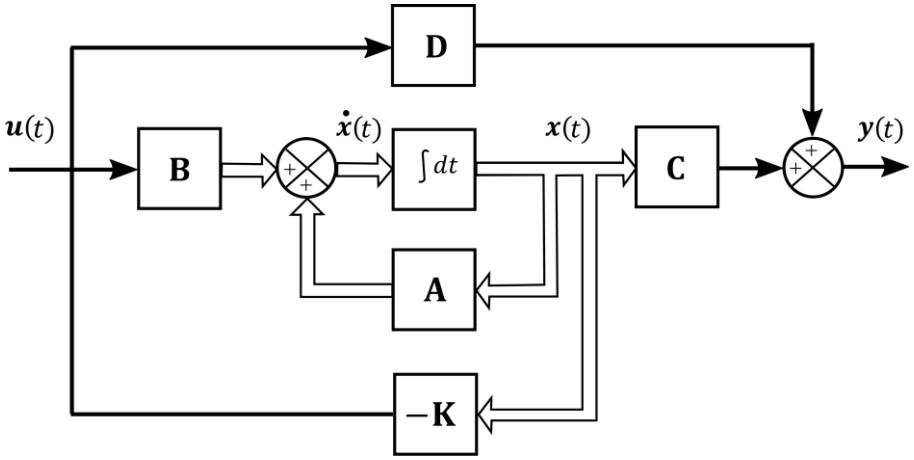

Fig. 1.5 Closed loop control system.

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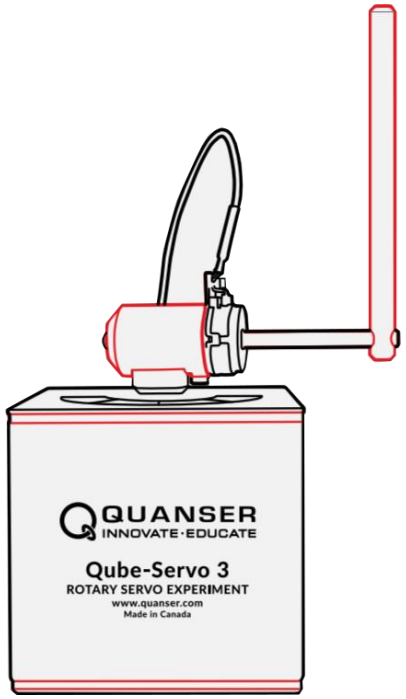
Part B. Regulator Control Design

Example: Inverted Pendulum



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5. Verify if the system is controllable using the controllability matrix.
6. Obtain the state feedback gain matrix **K**.

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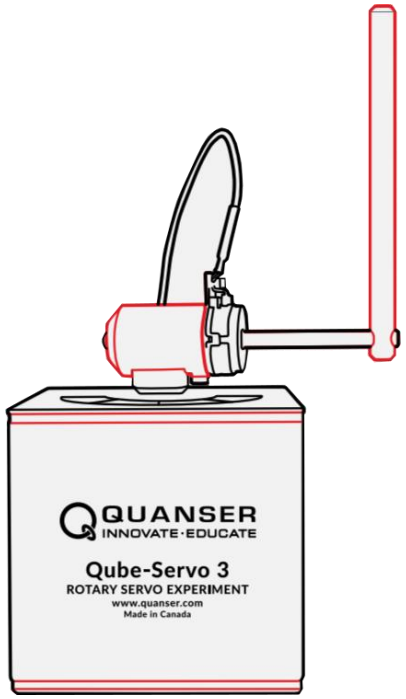
Part B. Regulator Control Design

Example: Inverted Pendulum



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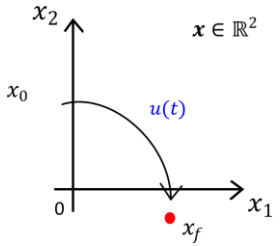

Kalman's Test of Controllability:

$$\boldsymbol{Q_c} = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B} \quad \cdots \quad \mathbf{A}^{n-1}\mathbf{B}]$$

Controllability Matrix

6. Obtain the state feedback gain matrix **K**.

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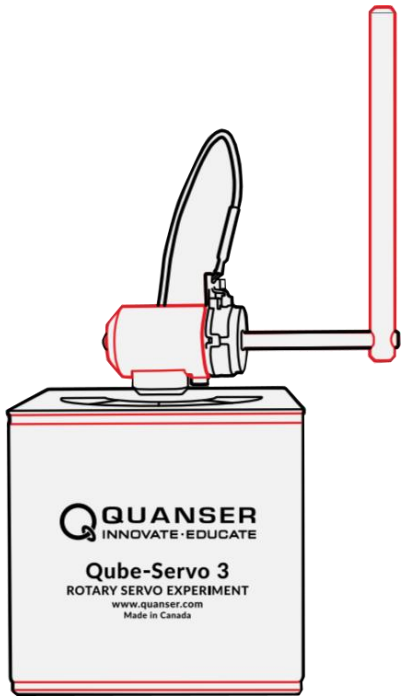
Practice 09

Part B. Regulator Control Design

Example: Inverted Pendulum



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Found $\begin{cases} p_1 = ? \\ p_2 = ? \end{cases}$ Add $\begin{cases} p_3 = 40 \\ p_4 = 45 \end{cases}$ $\Rightarrow$

desired polynomial

$$(s + p_1) * (s + p_2) * (s + p_3) * (s + p_4)$$

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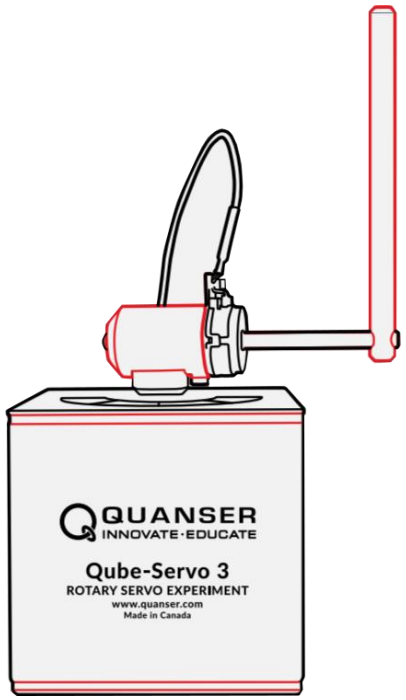
Practice 09

Part B. Regulator Control Design

Example: Inverted Pendulum



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Considering the system:

$$\dot{\boldsymbol{x}}(t) = \mathbf{A}\boldsymbol{x}(t) + \mathbf{B}\boldsymbol{u}(t)$$
$$\boldsymbol{y}(t) = \mathbf{C}\boldsymbol{x}(t) + \mathbf{D}\boldsymbol{u}(t)$$

With the control signal:

$$\boxed{\boldsymbol{u}(t) = -\mathbf{K}\,\boldsymbol{x}(t)}$$
$$\boxed{\dot{\boldsymbol{x}}(t) = (\mathbf{A} - \mathbf{B}\,\mathbf{K})\boldsymbol{x}(t)}$$

Design of the regulator system

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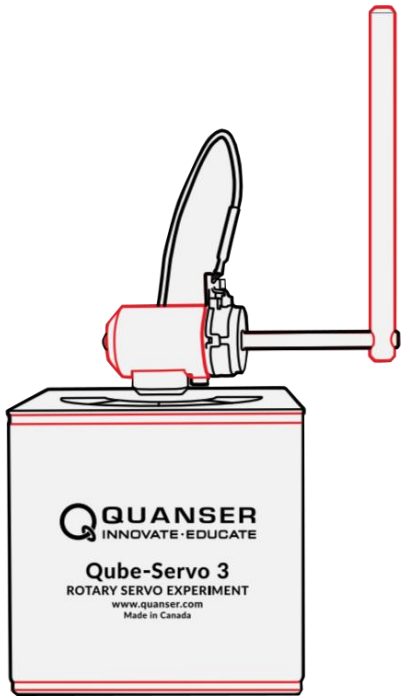
Part B. Regulator Control Design

Example: Inverted Pendulum



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Tecnológico de Monterrey

Thank you!

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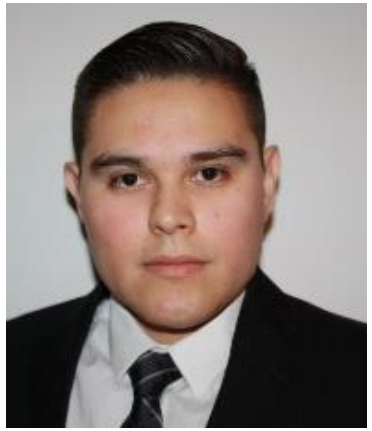
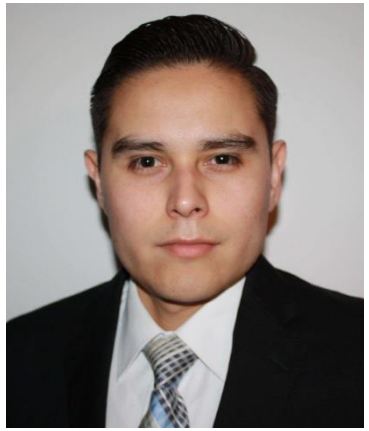
Carlos Sotelo
carlos.sotelo@tec.mx

Professor
David Sotelo
david.sotelo@tec.mx





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Oct 3rd, 2024